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MILITARY SPECIFICATION

HEATER, AIR, ELECTRICAL, FILTER UNIT, M3

This specification is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electric air heater for individual masks in a tank filter unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED. TEST METHOD STD NO. 101 - Test Procedures for Packaging Materials

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1275 - Characteristics of 28 Volt DC Electrical Systems in Military Vehicles

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-19-1782 - Heater, Air, Electric, Filter Unit, M3.

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-1782 - Heater, Air, Electric, Filter Unit, M3.

(Copies of standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

AMCA Standard 210 - Laboratory Methods of Testing Fans for Rating Purposes.

(Application for copies should be addressed to Air Movement and Control Association, Inc., 30 West University Drive, Arlington Heights, IL 60004.)

UNDERWRITERS' LABORATORIES, INC.

UL 499 - Electric Heating Appliances.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-19-1782 or on subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the item shall conform to the specifications and drawings listed on Drawing 5-19-1782 and subsidiary drawings.

3.2 Manufacture and assembly. The heater shall be manufactured and assembled as specified on Drawing 5-19-1782.

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3.2.1 Controller. The controller shall be removable from the heater for remote mounting and shall control the heater by suitable electrical cable extension. The temperature selector of the controller shall also be the heater "on-off" switch. In addition, the heater indicator light shall be integral with the controller and shall light during the heating cycle only. The controller assembly shall be totally inclosed and all electrical moveable contacts (relays, switches, etc.) shall be inclosed and watertight. The control circuit shall be designed so that all semiconductors, if used in the circuit, shall be capable of operating without damage from transient voltage up to 39 volts for 10 seconds.

3.2.2 Cable connector receptacles. The cable connector receptacles for the thermostat, power, and controller connections shall remain securely anchored when the male cable connector lead is pulled from the connection at a minimum angle of 60° from the longitudinal axis of the cable connector and when tested as specified in 4.4.4.1.

3.2.3 Component accessibility. The heating element, controller, and heater thermostat shall be easily accessible for maintenance and replacement.

3.2.4 Safety. The heater shall comply with the applicable paragraphs of UL 499 for internal wiring, electrical insulation, thermal insulation, and insulation resistance.

3.2.5 Screw threads. All screw threads shall be in accordance with FED-STD-H28.

3.3 Performance.

3.3.1 Efficiency. The heater, conditioned and maintained at -18° ± 2°C shall be capable of receiving air at -18° ± 2°C and at an airflow of 4.5 cubic feet per minute (cfm) minimum (corrected to standard conditions as defined in AMCA Standard 210, at its inlet and discharging it at a minimum of 49°C at its outlet within 15 minutes after starting heater when tested as specified in 4.4.4.2(a).

3.3.2 Resistance to airflow. With the heater operating at ambient room temperature the resistance to airflow shall not exceed a differential pressure of 2.5 millimeters (24.9 Pascals) of water when tested as specified in 4.4.4.2(b).

3.3.3 Heating element power consumption and operating range.

3.3.3.1 Heating element power consumption. With the heater operating as specified in 3.3.1 the heating element shall not consume more than 250 watts when tested as specified in 4.4.4.2(c)(1).

3.3.3.2 Heating element operating range. The heating element shall be capable of operating at ambient temperature within the range of 18 to 30 volts direct current when tested as specified in 4.4.4.2(c)(2).

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3.3.4 Thermostatic control. When tested in accordance with 4.4.4.2(d), with the intake air at ambient temperature and at a flow rate of 3.5 cfm, the controller and thermostat shall be capable of maintaining a temperature within a range of 2°C under the two following conditions:

(a) When the heater's temperature setting is adjusted above ambient temperature, but not greater than 8°C above ambient, and;

(b) When control knob is in the full clock-wise position.

3.3.5 Case surface temperature. With the heater operating at ambient room temperature the maximum surface temperature of the heater case shall not exceed 52°C when tested as specified in 4.4.4.2(e).

3.3.6 Electromagnetic emission and susceptibility. At all controller settings, the heater shall meet the requirements of CE01, CE03, and RE02 of MIL-STD-461. Emissions having a duration not exceeding 1 second, and recurring not more than once in 3 minutes, are exempt (see 4.3.4.1).

3.4 Leakage. the internal pressure of the heater, when raised to 30 cm (2.9 kilopascals) of water, shall not return to atmospheric pressure in less than 10 seconds. This requirement must be met after disassembly and reassembly of the end sections and when tested as specified in 4.4.4.3.

3.5 Operating life. With the heater operating at an airflow rate of 4.5 cfm (127.5 liter per minute), the controller shall be adjusted to an output temperature that results in the fastest cycle rate. The heater, operating at the aforementioned conditions, shall maintain the outlet temperature within a total range of 2°C for a continuous period of 250 hours when tested as specified in 4.4.4.4.

3.6 Resistance to environmental conditions. The heater shall pass the requirements of 3.3 after exposure to the environmental conditions specified in table I when tested as specified in table II.

3.7 Cold temperature operation. Immediately after being exposed to an environment of -43°C or colder for 4 hours, the heater shall operate (indicator light shall come on) when the control knob is turned to "on" position in accordance with 4.4.4.5.

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TABLE I. Environmental conditions.

Environmental condition	Test method 1/
Temperature shock	4.4.4.8
Humidity	4.4.4.9
Mechanical shock	4.4.4.10
Vibration	4.4.4.11
Salt fog	4.4.4.12
Fungus	4.4.4.13

1/ The heater shall not be operating during any of the tests.

TABLE II. Preproduction testing.

Group number	Requirement	Method of testing
I	Salt fog (3.6)	4.4.4.12
	Performance (3.3)	4.4.4.2
	Leakage (3.4)	4.4.4.3
II	Temperature shock (3.6)	4.4.4.8
	Cold temperature function (3.10)	4.4.4.5
	Humidity (3.6)	4.4.4.9
	Performance (3.3)	4.4.4.2
III	Mechanical shock (3.6)	4.4.4.10
	Vibration (3.6)	4.4.4.11
	Operating life (3.5)	4.4.4.4
	Performance (3.3)	4.4.4.2
IV	Fungus (3.6)	4.4.4.13
	Performance (3.3)	4.4.4.2
V	Electromagnetic emission and susceptibility (3.3.6)	4.3.4.1

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3.8 Compatibility with vehicular electrical system. When tested in accordance with 4.4.4.6, the heater circuitry shall withstand the voltage surges and spikes specified in MIL-STD-1275 for the following conditions:

(a) Fault free condition

(1) Combined generator-battery power supply
(2) Battery only

(b) Single fault condition

3.9 Identification marking. Each heater shall be permanently and legibly marked in accordance with MIL-STD-130 and shall include the voltage, wattage rating, and lot number.

3.10 Electrical grounding. The heating circuit shall not short nor ground when tested as specified in 4.4.4.7.

3.11 Workmanship. The heater shall be free from damage such as dented, cracked, or chipped surfaces, and foreign matter such as dirt, oily or viscous material.

3.12 Preproduction. Prior to the start of regular production, a preproduction sample of heaters shall be produced in accordance with this specification for examination and test (see 4.3). Any change in design made subsequent to preproduction sample approval shall necessitate performance of a new preproduction sample.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring

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that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2.1, 3.2.3, 3.2.4, 3.2.5 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 25 heaters shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production (see 6.2).

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive tests. Each heater in the preproduction lot shall be subject to examination and test. The sample of heaters shall be packaged in accordance with SPI P5-19-1782 (see 5.2), unless interplant shipment is required.

4.3.2.2 For destructive test. The preproduction sample of packaged heaters shall be tested for heat seal seam strength (4.4.4.15).

4.3.2.3 For preproduction tests. After the preproduction lot has been tested in accordance with 4.3.3.1, a sample of 10 heaters shall be selected at random from the lot and divided equally into five groups for testing in accordance with 4.3.3.2.

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive tests. Each packaged heater in the preproduction sample shall be examined in accordance with the classification of defects (4.4.3.2(b)) and tested for unit pack leakage (4.4.4.14). Each heater in the preproduction lot shall be examined and tested for all the requirements of the applicable drawings and all methods of testing defined in 4.4.4.2(a), 4.4.4.2(e), 4.4.3.2 and 4.4.4.6 of this specification.

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4.3.3.2 For destructive test. The preproduction sample of packaged heaters shall be tested for heat seal seam strength (4.4.4.15).

4.3.3.3 For preproduction tests. Sample heaters selected in accordance with 4.3.2.2 shall be tested in accordance with 4.3.4.

4.3.4 Tests. The heaters in each preproduction group shall be tested in the sequence and in accordance with the test methods specified in table II.

4.3.4.1 Electromagnetic emission and susceptibility. To determine conformance to 3.3.6, the heater shall at a minimum be tested at minimum, mid-range and maximum controller settings as specified in MIL-STD-462, as specified in table III.

TABLE III. Electromagnetic emission and susceptibility.

Requirement (MIL-STD-461)	Test (MIL-STD-462)	Frequency range	Application
CE01	CE01	30Hz - 20kHz 1/ narrowband	All power lines measured at connector
CE03	CE03	20kHz - 50MHz 2/ narrowband and broadband	All power lines measured at connector
RE02	RE02	14kHz - 10GHz 3/ narrowband	Steady rate
RE02	RE02	14kHz - 1GHz broadband	Steady rate

1/ kilo (k) Hertz (Hz)
2/ Mega (M)
3/ Giga (G)

4.3.5 Acceptance/rejection criteria. The preproduction lot shall comply with all the requirements of this specification when examined and tested in accordance with 4.3.3. The acceptance number is zero. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the heaters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. Each lot shall be identified and controlled in accordance with MIL-STD-1168.

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4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. For 4.4.3.2(a), major defect 101, inspection level S-3 shall be used. Unless interplant shipment is required, the packaged heaters shall be tested for unit pack leakage (4.4.4.14) using inspection level S-3.

4.4.2.2 For destructive test. Unless interplant shipment is required, the packaged heaters shall be tested for heat seal seam strength (4.4.4.15) using inspection level S-3.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample heaters shall be examined and tested in accordance with the classification of defects and MIL-STD-105. Each sample packaged heater shall be tested for unit pack leakage using inspection level S-3 and an AQL of 4.0 percent.

4.4.3.1.2 For destructive test. The sample of packaged heaters shall be tested for heat seal seam strength (4.4.4.15), using inspection level S-3 and an AQL of 4.0 percent.

4.4.3.2 Classification of defects.

(a) Heater, air, electric, filter unit, M3 (Dwg E5-19-1782).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Cable connector receptacles	4.4.4.1
102	Resistance to airflow	4.4.4.2(b)
103	Heating element power consumption	4.4.4.2(c)(1)
104	Heating element operating range	4.4.4.2(c)(2)
105	Thermostatic control	4.4.4.2(d)
106	Leakage	4.4.4.3
107	Electric grounding	4.4.4.7
108	Mounting holes incorrect or incorrectly located (Detail c)	
109	Thread size of outlet and inlet openings incorrect	
110	Pipe plugs missing or incorrect	
111	Electrical lead markers missing	
112	Marking missing or incorrect	
113	Workmanship (3.11)	

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Minor: AQL 2.5 percent defective

201	Marking illegible
202	Protective finish missing

(b) Packaging inspection (SPI P5-19-1782).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Item not completely clean/dry
102	Unit pack assembly incorrect
103	Base incorrect
104	Liner incorrect
105	Interior container (box) and closure incorrect
106	Barrier and closure incorrect
107	Marking missing, incorrect or illegible

4.4.4 Tests.

4.4.4.1 Cable connector receptacles. The cable connector receptacles shall remain securely anchored when the male cable connectors are pulled loose from their respective receptacle at the minimum specified angle. This test shall be performed at ambient room temperature.

4.4.4.2 Performance.

(a) **Efficiency.** The test shall be conducted at -18° ± 2°C and shall consist of two test "hook-ups" - one using the end inlet and outlet ports and one using the side inlet and outlet ports of the heater. The heater shall be conditioned for 24 hours at -18°± 2°C prior to testing each "hook-up". Air, at -18° ± 2°C and a minimum of 4.5 cfm, shall pass through a calibrated metering device (rotometer or gas meter) and then through the heater. Two calibrated thermocouples shall be installed in the system, one at the inlet and one at the outlet port, to measure the inlet and outlet air temperature. The thermocouples shall be installed approximately one inch from their respective ports. The heater shall be connected to a suitable power source capable of producing 27.5 volts direct current (vdc). The heater shall be energized with 27.5 vdc. Air entering the inlet port of the specified temperature and flow rate shall be discharged through the outlet port at a minimum temperature of 49°C within 15 minutes after starting the heater: however, no heater shall be rejected unless it has been operated for 15 continuous minutes.

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(b) **Resistance to airflow.** The pressure differential shall be measured with a suitable manometer across the heater at the conditions specified in 4.4.4.2(a) except that the temperature shall be at ambient room temperature. The differential pressure shall not exceed the maximum specified.

(c) **Heating element power consumption and operating range.**

(1) **Heating element power consumption.** The heater shall be operated as specified in 3.3.3.1 except that a voltmeter and ammeter shall be appropriately connected to the electrical circuit. The voltage and amperage measurements shall be taken and the heating element rating determined. The rating shall not exceed the maximum specified.

(2) **Heating element operating range.** Operations of the heating element shall be verified at the upper and lower required direct current voltages by the indicator light coming on when the control knob is turned on.

(d) **Thermostatic control.** Thermocouples shall be installed in the heater to measure the inlet and outlet temperature of the air. The thermostatic control shall be adjusted to two settings specified in 3.3.4(a) and 3.3.4(b) and the outlet air temperature shall be allowed to stabilize. After stabilization the heater shall maintain the temperature of the outlet air of each setting within the specified tolerance for 10 minutes without further controller adjustment.

(e) **Case surface temperature.** The heater shall be tested as specified in 4.4.4.2(a) except that six additional calibrated thermocouples shall be attached to six scattered points, at least one of which shall be attached to the outlet boss, on the heater surface. During operation of the heater, no case surface temperature measurement shall exceed the maximum specified.

4.4.4.3 Leakage. Three of the four openings of the heater shall be sealed (capped or plugged). A pressurizing source shall be connected to the fourth opening. A manometer shall be connected to the system not farther than two inches from the heater opening and a valve shall be installed in the system between the pressurizing source and the manometer. The heater shall be pressurized to 30 centimeters (2.9 kilopascals) of water and the valve shall be closed. The inside pressure must not return to atmospheric pressure in less than the minimum specified.

4.4.4.4 Operating life. Ambient room temperature air shall be passed through a calibrated metering device (rotometer or gas meter) through the heater, and exhausted outside the test area so that the heated exhaust will have no influence on the ambient room temperature. Thermocouples and current source shall be as specified in 4.4.4.2(a). The thermostatic controller and the airflow rate shall be adjusted to obtain an outlet temperature that results in the fastest heater cycle rate and maintain that temperature

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within a range of 2°C for the minimum specified period of time. The heater indicator light on the controller shall be lit while the heater is heating for each cycle.

4.4.4.5 Cold temperature operation. After 4 hours at -43°C or colder, the control knob shall be turned to the "on" position. The heater shall function.

4.4.4.6 Compatability with vehicular electrical system. The heater shall be tested with test equipment capable of measuring the performance envelopes (only upper voltage limits apply) defined by 3.8 and associated requirements and test parameters of MIL-STD-1275.

4.4.4.7 Electrical grounding. While the heater is "on" (indicator light kit) rotate the inlet and outlet ports 180° each side of the vertical position. The heater shall not malfunction due to grounding or shorting of internal wiring.

4.4.4.8 Temperature shock. The heater shall be subjected to temperature shock as follows:

(a) Place the test item in a high-temperature chamber, and raise the chamber temperature to 71°C (160°F). Maintain for a period of not less than 4 hours or until the test item stabilizes.

(b) Transfer the test item, within 5 minutes, to a cold chamber with an internal temperature of -57°C (-70°F). Maintain for a period of not less than 4 hours or until the test item stabilizes.

(c) Return the test item, within 5 minutes, to the high-temperature chamber maintained at 71°C (160°F). Maintain for a period of not less than 4 hours or until test item stabilizes.

(d) Repeat steps (b) and (c) above.

(e) Repeat step (b) above.

(f) Return the test item to standard ambient conditions and stabilize.

4.4.4.9 Humidity. The heater shall be subjected to humidity as follows:

(a) Dry the test item at 54°C (129°F) for 24 hours.

(b) Condition the test item at 23°C (73°F) and 50 ± 10 percent relative humidity for 24 hours.

(c) Raise the internal chamber temperature to 30°C (86°F) and the relative humidity to 94 ± 4 percent.

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(d) Subject the test item to five continuous 48-hour cycles in accordance with figure 1.

(e) Condition the test item for 24 hours at 23°C (73°F) and 50 ± 10 percent relative humidity.

4.4.4.10 Mechanical shock. Three shocks in each direction shall be applied along three mutually perpendicular axes of the test item (total of 18 shocks). The shock pulse configuration shall conform to figure 2.

4.4.4.11 Vibration. The test item shall be vibrated along each axis in accordance with curve W of figure 3. Test time schedule (per axis) shall be as follows:

(a) Sinusoidal cycling time: 30 min/1000 miles.

(b) Maximum cycling time: 3 hours.

(c) Sweep time: 15 min (5-500-5 Hz). Sweep time shall be increased by 3 minutes if test frequencies go to 2 Hz.

4.4.4.12 Salt fog. Exposure to salt fog shall be in accordance with method 509.2, procedure I, MIL-STD-810. Salt deposits may be removed with distilled water.

4.4.4.13 Fungus. Exposure to fungus shall be in accordance with method 508.3 of MIL-STD-810.

4.4.4.14 Unit pack bag leakage. The leakage of the unit pack bag containing the heater shall be determined by subjecting the sample to the hot water technique, method 5009.2 of FED. TEST METHOD STD. NO. 101.

4.4.4.15 Heat seal seam. The heat seal seam of the unit pack bag shall be determined by subjecting the sample to FED. TEST METHOD STD. NO. 101, method 2024, except that the static load shall be 45 ounces ± 5 ounces, and a 5 percent reduction shall be permitted when room temperature in test area exceeds 90°F.

5. PACKAGING

5.1 Interplant shipment (see 6.4). Military packaging as specified in 5.2 is not applicable for interplant shipments. The heaters shall be packaged to provide adequate protection from physical damage, and marked to identify the filters, and provide any other required data during shipment from the source to the first receiving activity.

5.2 Repair or replacement parts. For all shipments, other than interplant shipments (see 6.4), packaging (preservation, unit packing, packing and marking), shall be as specified in SPI P5-19-1782.

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6. NOTES

6.1 Intended use. The M3 electric air heater covered by this specification is a component of the gas-particulate filter units M13 and M13A1 and is used to warm respiratory air supplied to individual masks.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of packing required
(c) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Subject term (key word) listing.

Air heater
Electric heater
Filtered air
Respiratory air
Respiratory mask
Tank air heater
Temperature controller

6.4 Interplant shipment. Packaging and marking for interplant shipment is for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are shipments from a vendor or subcontractor or a prime contractor, or between contractors and subcontractors, or from a vendor or contractor to a military arsenal, plant, or other activity for evaluation, immediate use, or further processing as specified in the applicable contract.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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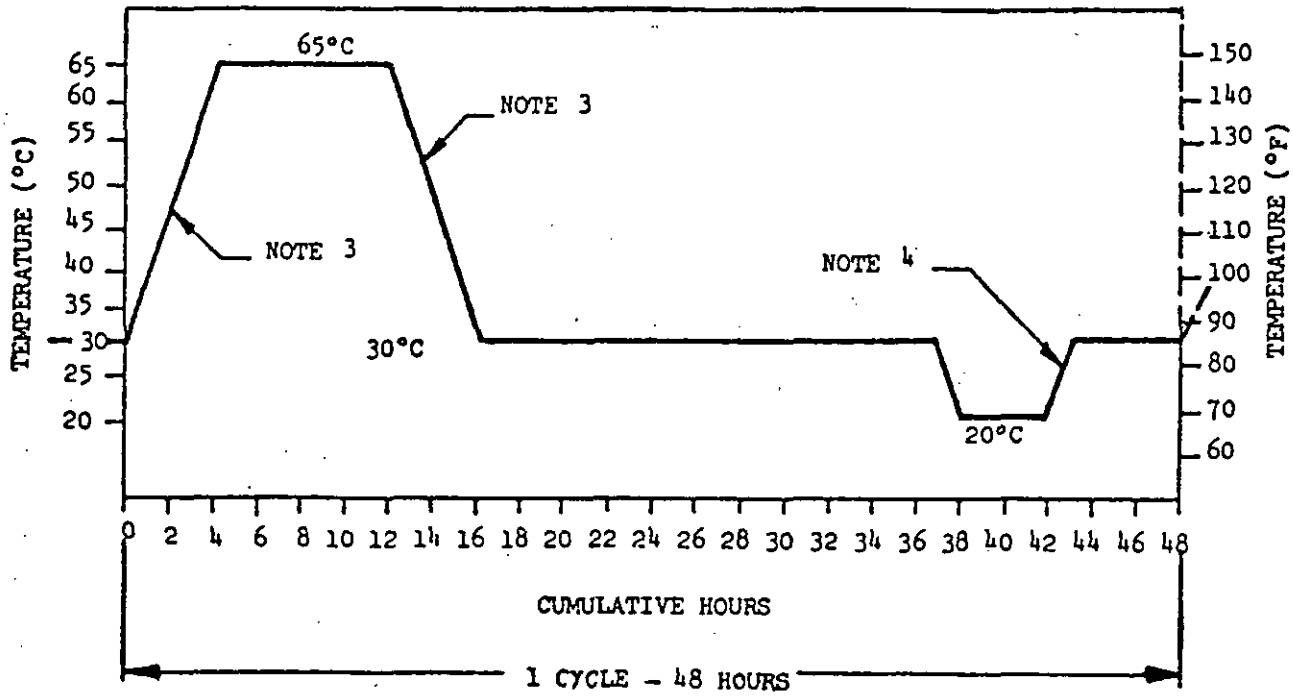
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NOTES:

1. Tolerance during temperature change shall be not greater than 3°C (5°F).

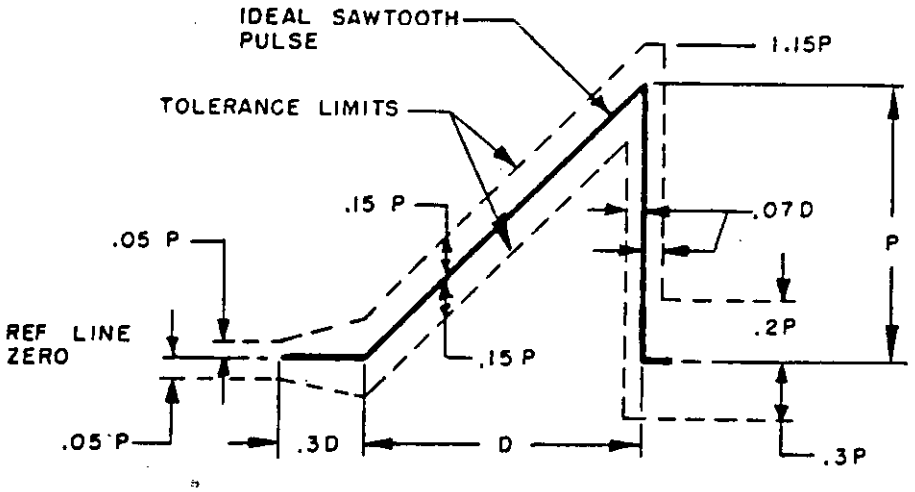
2. Relative humidity shall be maintained at 94 ±4 percent at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85 percent.

3. Rate of temperature change between 30° and 65°C (86° and 149°F) shall be not less than 8°C (14.4°F) per hour.

4. The temperature increase in this portion of the curve shall be not less than 10°C (18°F).

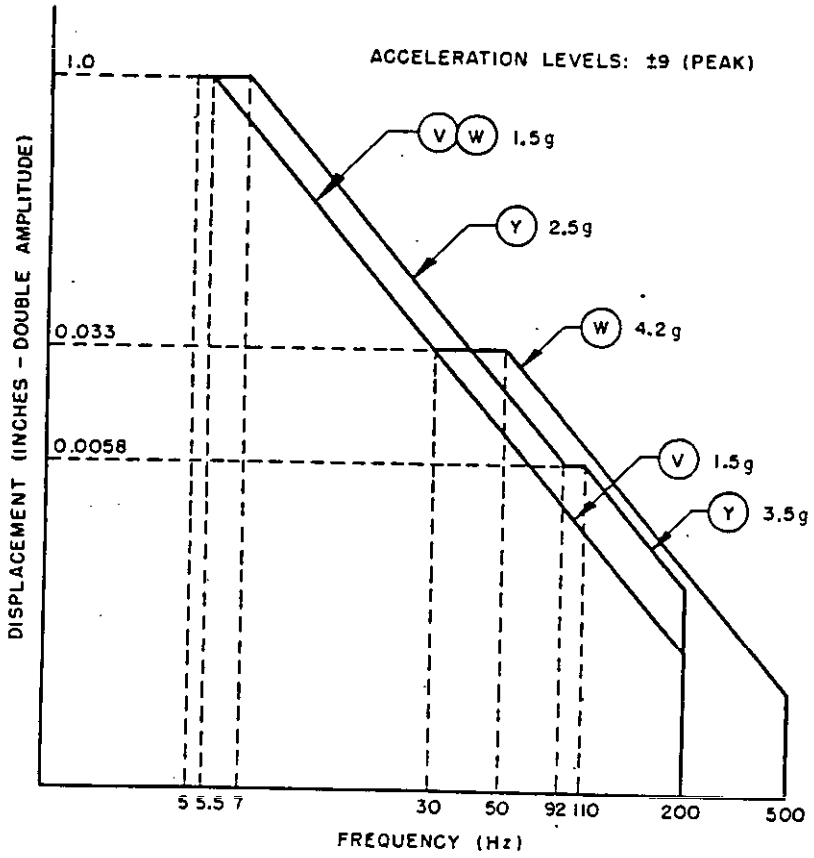
FIGURE 1. Humidity cycle.

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P = 30 g (g = 980.6 cm/S^2)
D = 11 milliseconds

FIGURE 2. Shock pulse configuration.



Note: All curves shall be extended to 2 Hz when test item resonances below 5 Hz are expected

(g = 980.6 cm/S^2)

FIGURE 3. Vibration test curves.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-51191D(EA)	HEATER, AIR, ELECTRICAL, FILTER UNIT, M3

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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